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The influence of confining pressure on the shear strength of compacted rockfill

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Large-scale drained triaxial compression tests have been carried out on heavily compacted samples of different types of rockfill over a range of confining pressures. The Mohr failure envelopes show pronounced curvature, particularly at low stresses. An assessment has been made of the significance of this for the stability of rockfill slopes where generally the shear strength of the rockfill at low stresses is of major importance. It is concluded that in many instances rockfill embankments could be built with slopes much steeper than those currently adopted and still have an adequate factor of safety against stability failure.

Des essais de compression triaxiale drainés à grande échelle ont été réalisés sur des échantillons fortement compactés de différents types de remblai rocheux, suivant toute une gamme de pressions d'éntreinte latérale. Les courbes limites de Mohr présentent une courbure prononcée, notamment aux faibles contraintes. On etabli la signification de ce résultat vis-à-vis de la stabilité des talus de remblai rocheux dont la résistance au cisaillement pour de faibles contraintes est généralement predominante. On en a conclu que, dans bien des cas, il était possible de construire des talus de remblai rocheux avec une pente beaucoup plus raide que celle des talus couramment construits, le coefficient de sécurité relatif au glissement restant adequat.

INTRODUCTION

Great efforts have been made over the past 20 years to measure the shear strength of the well-graded and heavily compacted rockfills now typically used in embankment construction. A major difficulty in this has been the size of rock fragments placed in rockfill embankments and this had led to the building of large-scale laboratory testing facilities. A triaxial cell was built in Mexico, in connection with the design of the 148 m high Infiernillo Dam, in which samples 1·1 m dia. by 2·5 m high could be tested using confining pressures up to 2500 kN/m^2 and with axial loads up to 1500 t (Marsal *et al.*, 1965; Marsal, 1967). Large-scale rockfill testing equipment was also built at the University of California to test samples 0·9 m dia. by 2·3 m high in triaxial compression with confining pressures up to 5000 kN/m^2 and axial loads up to 1800 t (Marachi *et al.*, 1969; Marachi, Chan and Seed, 1972).

It has been customary to express the shear strength, measured in a drained test with a particular confining pressure in terms of

$$\phi' = \sin^{-1} \frac{(\sigma_1'/\sigma_3')_f - 1}{(\sigma_1'/\sigma_3')_f + 1} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (1)$$

where ϕ' is the angle of shearing resistance and $(\sigma_1'/\sigma_3')_f$ is the maximum principal stress ratio, i.e. it is assumed that the cohesion (c') is zero.

When the rockfill used in the field contains fragments too large for a laboratory triaxial sample, a method of scaling down the field grading must be adopted. This will often be the case even with the large-scale equipment previously described, as the field grading may include rock fragments with dimensions of one metre or more. Lowe (1964) suggested the use of a laboratory grading which would plot as a line parallel to the field grading on a semi-logarithmic

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particle size distribution graph. Marachi *et al.* (1969) described tests in which for each of three rockfills, three parallel gradings were tested using different size samples so that the ratio of sample diameter to maximum particle size was always 6.[1] For all three materials it was found that the measured angle of shearing resistance decreased slightly as the maximum particle size increased; typically a sample with a maximum particle size of 150 mm had an angle of shearing resistance about 4° smaller than the same material with a maximum particle size of 12 mm. However, tests carried out by Tombs (1969) and Charles (1973) at Imperial College on 300 mm dia. samples of a number of different types of rockfill with a maximum particle size of 75 mm showed angles of shearing resistance typically about 2° greater than for samples with a maximum particle size of 10 mm. It is difficult to assess with certainty the relatively minor influence of maximum particle size on the angle of shearing resistance due to a number of other factors which have a significant effect on the measured angle of shearing resistance. All three materials tested by Marachi *et al.* (1969) showed a marked decrease in angle of shearing resistance with increased confining pressure. Typically the angle of shearing resistance measured in a test on a particular material at a confining pressure of 200 kN/m^2 was 10° greater than the angle measured in a test with a confining pressure of 4500 kN/m^2. It was also found that at low stresses the initial porosity had a major effect on the shear strength. At a confining pressure of 200 kN/m^2, an increase of 1% in initial porosity could reduce the angle of shearing resistance by 0·5°.

The stresses in rockfill structures will in general approximate to plane strain conditions rather than the triaxial situation. Consequently, the laboratory work in Mexico was extended to the testing of rockfill samples 0·7 m by 0·75 m in cross-section and 1·8 m high under plane strain conditions (Marsal *et al.*, 1967). When plane strain tests are compared with triaxial tests at the same minor principal stress, it is found that the principal stress ratio at failure is significantly greater in the plane strain tests (Marsal, 1973).

Considerable emphasis has been placed on triaxial tests carried out at high confining pressures in the large-scale rockfill testing in Mexico and California. Confining pressures up to 2500 kN/m^2 have been used in Mexico and up to 4500 kN/m^2 in California. However, in most field cases the shear strength of rockfill at comparatively low stresses is much more important. The maximum normal effective stress on a critical failure surface in a 50 m high rockfill embankment would be unlikely to be greater than 400 kN/m^2. Much of the interest in the shear strength of rockfill should be concentrated on its behaviour at comparatively low confining pressures. The purpose of the present investigation has been to measure the influence of confining pressure on the angle of shearing resistance for a number of rockfills at low and medium confining pressures and to assess its significance for the stability of rockfill structures. Availability of equipment meant that triaxial tests only could be carried out. The sample size diameter (230 mm) limited the maximum particle size to 38 mm. The tests have been carried out on heavily compacted samples of rockfills at their placement moisture content to simulate field conditions in modern rockfill embankment construction as closely as possible. The effect of variations in grading, maximum particle size, initial density and moisture content have not been examined.

LABORATORY TESTING PROGRAMME

Drained triaxial compression tests have been carried out on samples 0·23 m in diameter by

[1] Marachi *et al.* (1969) concluded from data presented by Holtz and Gibbs (1956) that if the material gradation is such that the proportion of particles in the maximum sieve size range is less than 30%, then a ratio of sample diameter to largest particle size of 6 or more will ensure that the sample size does not affect the test results. In some investigations a ratio of 4 has been used.

Table 1. Properties of parent rock materials

Rock type	Specific gravity	Saturated moisture content: %	Particle shape		Strength indices: MN/m²			
			d_1/d_2	d_2/d_3	I_1		I_2	
					Dry	Sat.	Dry	Sat.
A. Sandstone	2·60	4	1·4	1·5	138	133	10	5
B1. Slate	2·81	2	1·6	2·0	102	75	21	13
B2. Slate	2·71	10	1·5	1·7	35	22	3·8	2·4
C. Basalt	2·80	3	1·5	1·3	219	125	19	15

Table 2. Average initial properties of triaxial samples

Rock type	Dry unit weight: kN/m³	Moisture content: %	Porosity: %	Degree of saturation: %
A. Sandstone	20	7	20	70
B1. Slate	21	5	25	50
B2. Slate	18	16	32	90
C. Basalt	21	6	25	55

0·5 m high in a steel triaxial cell manufactured by Engineering Laboratory Equipment Ltd. The maximum axial load that could be applied by the load frame was 500 kN. The axial load was measured by a load cell inside the triaxial cell. Two load cells were used in the series of tests, one with a range 0 to 100 kN and the other 0 to 500 kN. The confining pressure was applied by a conventional oil–water constant pressure system. As the rockfill samples were not saturated, the volume of water and air expelled from the sample (or entering the sample due to dilation) did not give a very accurate indication of the volume changes of the sample. It was considered preferable to monitor the volume of water entering or leaving the cell. This volume, when corrected for volume changes due to volume of the ram and the expansion of the steel cell, gave a reliable measurement of the volume change of the sample. Samples were well graded and dense so that large errors due to membrane penetration were unlikely even during the application of the confining pressure and certainly were not a problem during the shear stage of the test. Automatic logging of all measurements was introduced during the series of tests and this necessitated the construction of a new volume change measuring device (Watts, 1980). Most of the tests were carried out at an axial strain rate of 1·4% per hour.

Four rockfill materials were tested. These are described in Tables 1 and 2. In every case the tested material had a maximum particle size of 38 mm and the sample was formed in a 0·23 m dia. split mould. Samples were built in 75 mm deep layers and each layer was compacted for one minute with an electric vibrating hammer fitted with 145 mm dia. tamping foot. The hammer and compaction procedure are described in British Standard 1377, Test 14. The rockfill was placed and compacted at its natural moisture content. The degree of saturation of the samples after compaction varied between 50% and 90%.

Crushing tests were carried out on rock fragments using the procedure described by Hobbs (1963). Irregular lumps of rock weighing about 30 g (Fig. 1) were crushed between hardened steel end platens covered with carbon paper and graph paper. The load on the rock specimen was increased until a split appeared. Two strength indices are quoted in Table 1.

$$I_1 = L/a$$

$$I_2 = L/d^2$$

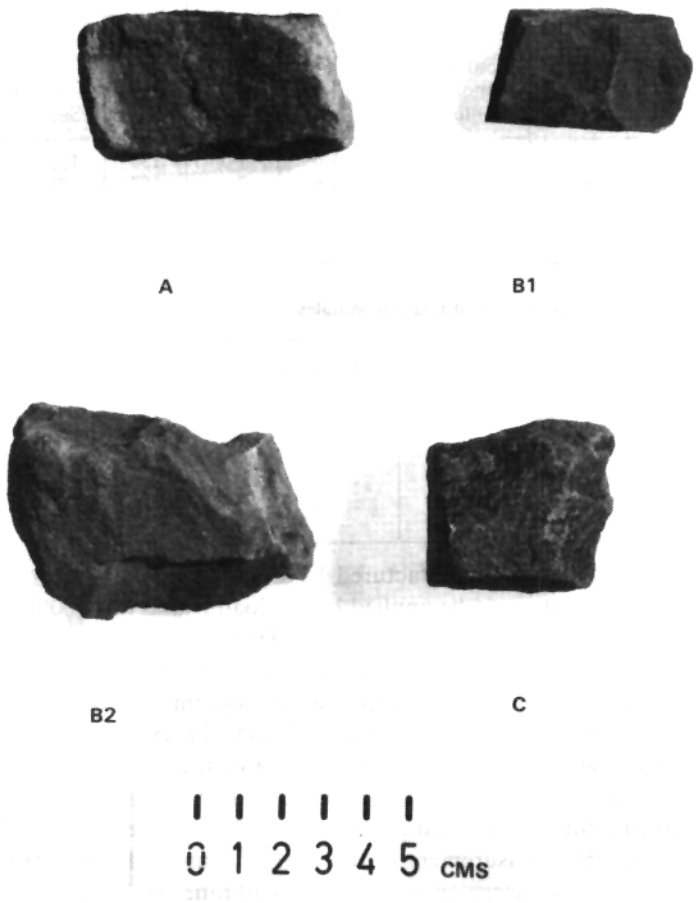


Fig. 1. Fragments of the four rockfills listed in Table 1

where L is the maximum load, a is the average area in contact at failure as measured by the carbon and graph paper and d is the dimension of the rock fragment in the direction of load application.

Strength indices are quoted for each rock material in both a dry and a saturated (surface dry) condition. The dimensions of the rock fragments were measured and the ratios of length to width (d_1/d_2) and width to depth (d_2/d_3) are quoted in Table 1. The saturated moisture content of the fragments was also measured. The two strength indices give a different picture of the relative strengths of the different rockfills. All four materials show a significant decrease in strength (of up to 50%) when saturated. This effect was discussed by Terzaghi (1960). The triaxial test conditions were such that the larger rock fragments in the triaxial samples would be close to saturation and therefore the strength indices of the saturated rock fragments are the more relevant parameters. The slate (B2 in Fig. 1) is the weakest of the materials and also has the highest moisture content. The index I_2, which is directly related to the load at which rock fragments fracture, suggests that the basalt (C) was the strongest of the four saturated materials. The ratios of dimensions show that the slate (B1) had fragments that were most

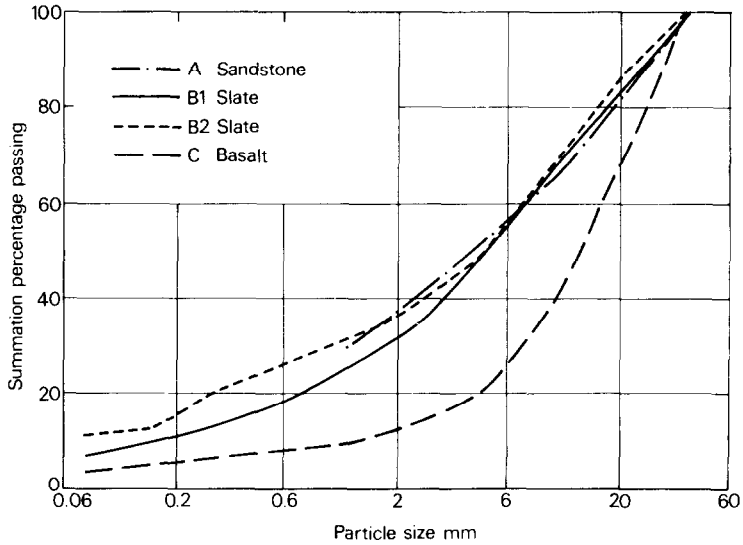


Fig. 2. Typical gradings of rockfill triaxial samples after testing

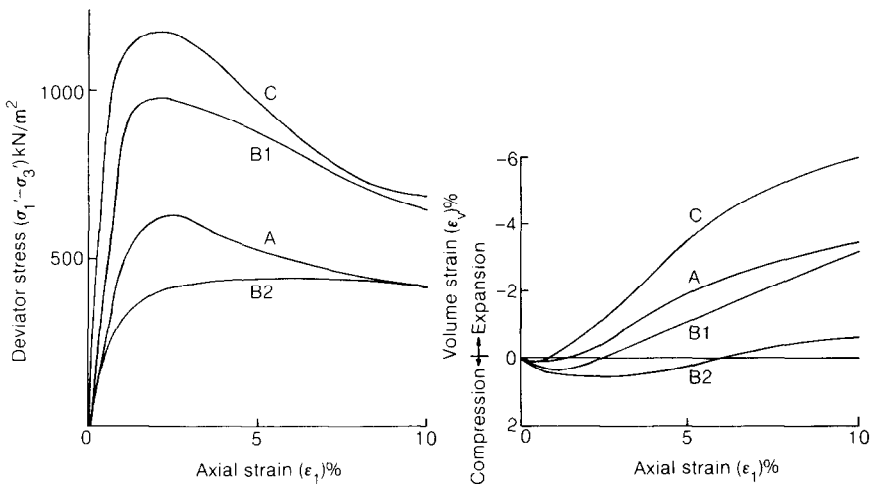


Fig. 3. Drained triaxial compression tests on heavily compacted rockfills with $\sigma_3' = 100$ kN/m^2

markedly platy in shape. The four rockfills show a wide range of both strength and shape of the parent rock materials. The weak, soft slate (B2) might be rejected as unsuitable for rockfill material in many civil engineering works whereas the hard igneous basalt (C) is at the opposite end of the range of rockfill materials. It is considered that the three rockfills, sandstone (A), slate (B1) and basalt (C), are reasonably representative of what would be considered good rockfill material for most civil engineering applications.

Typical gradings of the four rockfill materials after testing are shown in Fig. 2. The sandstone (A) and the two slates (B1 and B2) have broadly similar gradings. The basalt (C) has a more uniform grading with a smaller fines content. Further information about the properties of the rockfill samples tested in the triaxial test programme is presented in Table 2. The

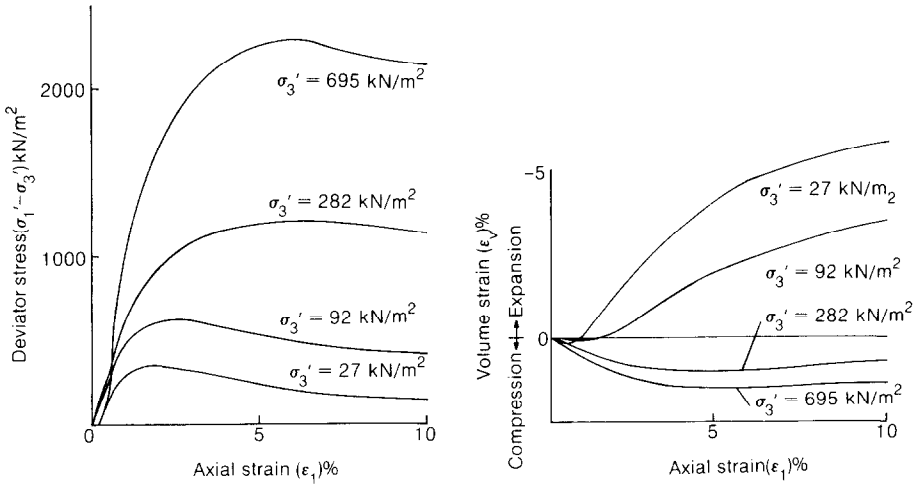


Fig. 4. Drained triaxial compression tests on heavily compacted sandstone rockfill (A) at various confining pressures

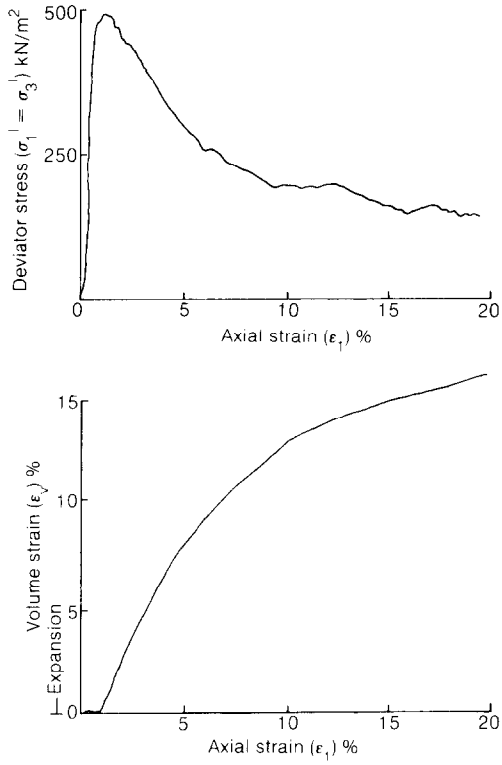


Fig. 5. Drained triaxial compression test on heavily compacted basalt rockfill (C) at low confining pressure (σ_3' = 27 kN/m^2)

sandstone (A), slate (B1) and basalt (C) have porosities in the range 20 to 25% and degrees of saturation of 50 to 70%. The relatively low density of the slate (B2) may be associated with its high moisture content and degree of saturation.

The results of triaxial tests on the four rock materials at a confining pressure of 100 kN/m^2 are shown in Fig. 3 as plots of deviator stress and volume strain versus axial strain. The basalt rockfill (C) has the greatest deviator stress at failure. It also shows the most marked reduction in deviator stress after failure. The axial strain at failure is close to 2% for rockfills A, B1 and C. The slate (B2) alone shows no peak behaviour. The basalt (C) shows the maximum and the slate (B2) the minimum dilatancy rate (expressed as $-d\varepsilon_v/d\varepsilon_1$). Although the slate (B1) has a greater deviator stress at failure than the sandstone (A), the sandstone has a slightly greater rate of dilation. There is some correlation between the maximum deviator stress and the strength of the rock fragments as measured by the strength index I_2 on saturated rock fragments (Table 1).

The results of triaxial tests on the sandstone rockfill (A) at four different confining pressures are shown in Fig. 4. At a very low confining pressure ($\sigma_3' = 27$ kN/m^2) the dilation is marked whereas at higher pressures ($\sigma_3' = 282$ kN/m^2 and $\sigma_3' = 695$ kN/m^2) there is little dilation. The axial strain at which failure occurs increases as the confining pressure increases. At the lowest pressure the failure strain is under 2%, at the highest pressure it is 6%.

It is clear that the high ratios of principal stress at failure measured with samples at low confining pressure must be sensitive to any excess pore pressure or suction in the sample. The strongest of the fills, the basalt (C) was tested at the lowest confining pressure ($\sigma_3' = 27$ kN/m^2) on a sample which had been 'saturated' by passing water through it before testing. The test was carried out at a slow rate of axial strain of 0·05% per hour. The results of this test are plotted in Fig. 5. Despite these precautions which ensured that no suctions developed, an extremely high value (19) was recorded for $(\sigma_1'/\sigma_3')_f$ and the dilation rate $(-d\varepsilon_v/d\varepsilon_1)_f$ was over 2.

INFLUENCE OF CONFINING PRESSURE ON SHEAR STRENGTH OF COMPACTED ROCKFILL

It is generally accepted that values of ϕ' measured in drained tests on particulate materials increase with a decrease of confining pressure. Drained triaxial compression tests by Skinner (quoted by Bishop, 1966) on dense saturated Ham River sand showed $(\sigma_1'/\sigma_3')_f$ increasing from 3·1 in a test with $\sigma_3' = 6900$ kN/m^2 to 4·3 in a test with $\sigma_3' = 690$ kN/m^2. Vesic and Clough (1968) similarly showed $(\sigma_1'/\sigma_3')_f$ increasingly significantly as the mean normal stress of tests on dense samples of a number of sand and gravel materials was reduced from 10 000kN/m^2 to 100 kN/m^2. Ponce and Bell (1971) carried out drained triaxial compression tests at low pressures on very dense saturated samples of a uniform sand; $(\sigma_1'/\sigma_3')_f$ was found to vary from 4·3 in a test with $\sigma_3' = 35$ kN/m^2 to over 8 in a test with $\sigma_3' = 1{\cdot}5$ kN/m^2. Billam (1971) tested dense samples of a limestone sand over a wide range of confining pressure. With $\sigma_3' = 10\,000$ kN/m^2, $(\sigma_1'/\sigma_3') = 4$; with $\sigma_3' = 3$ kN/m^2, $(\sigma_1'/\sigma_3') = 12{\cdot}5$. Penman (1953) reported a similar trend in the behaviour of silt.

The results of many large-scale tests carried out on a wide range of granular fill materials were reported by Marsal (1973). A series of drained triaxial compression tests was carried out on each material over a wide range of confining pressures. It was found that the principal effective stress ratio at failure $(\sigma_1'/\sigma_3')_f$ was much greater for tests carried out at low confining pressure (σ_3') than for tests carried out with higher values of σ_3'. In Fig. 6 the large hatched area indicates the range of Marsal's results, which included tests on sand and gravel as well as rockfill and tests with different amounts of initial compaction. Figure 6 (small hatched area) also shows results presented by Marachi *et al.* (1972) of large-scale drained triaxial compression

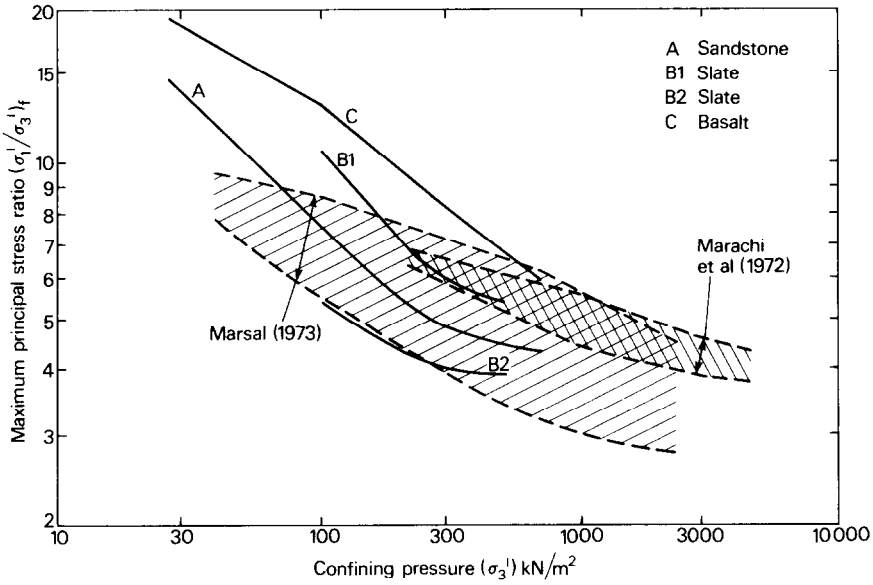


Fig. 6. Influence of confining pressure (σ_3') on maximum principal stress ratio $(\sigma_1'/\sigma_3')_f$

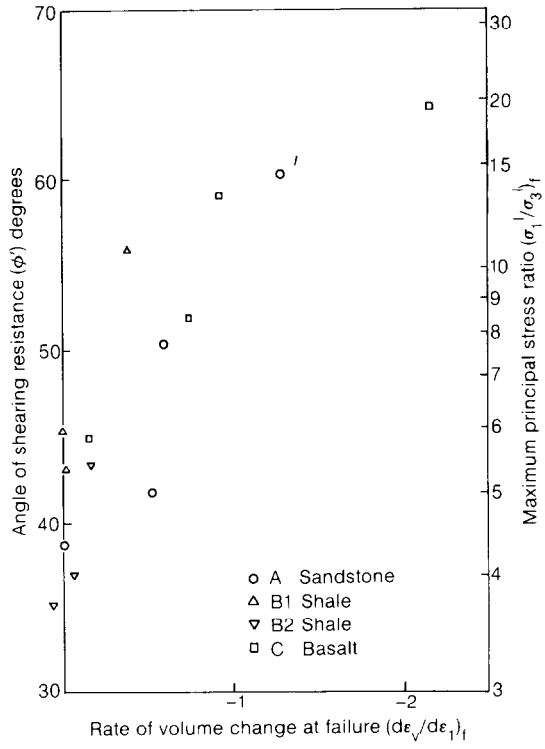


Fig. 7. Influence of rate of volume change at failure $(d\varepsilon_v/d\varepsilon_1)_f$ on angle of shearing resistance (ϕ')

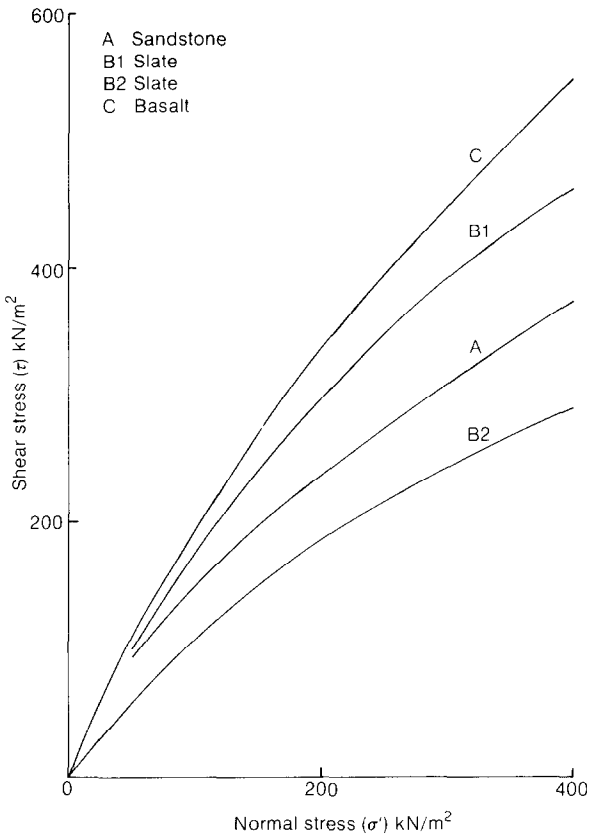


Fig. 8. Mohr failure envelopes for rockfills at low and medium confining pressures

tests on three well-compacted granular fills. Both series of tests were carried out on samples about one metre in diameter. The results of the tests on samples 0·23 m in diameter carried out at the Building Research Station (BRS) and described in the previous section are also plotted.

Although at high confining pressures the different series of tests show a similar range of $(\sigma_1'/\sigma_3')_f$, at low confining pressures the BRS tests indicate higher values of this ratio. Unfortunately, Marachi *et al.* (1972) do not present results for low values of σ_3'. Marsal gave only three results for rockfill materials tested with $\sigma_3' = 40$ kN/m^2 and the initial relative density of these three samples was not particularly high (45 to 70%). At higher relative densities, greater values of $(\sigma_1'/\sigma_3')_f$ would almost certainly have been obtained and consequently it is not considered that the results of the various series of tests are in serious conflict.

Figure 7 illustrates how the very high ratios of $(\sigma_1'/\sigma_3')_f$ obtained in the BRS tests on rockfills at low confining pressures are associated with high rates of dilatancy. Indeed at a very low confining pressure ($\sigma_3' = 27$ kN/m^2), $(-d\varepsilon_v/d\varepsilon_1)_f$ is considerably greater than 1 for the sandstone and basalt rockfills. Marsal (1973) has shown that the reduction in $(\sigma_1'/\sigma_3')_f$ (and also the reduction in dilatancy) at higher confining pressures is linked to particle breakage. While noting that the suppression of dilatancy was accompanied by particle fracture in natural sands and rockfill, Bishop (1971) pointed out that the effect occurs, although to a lesser extent, in samples of steel shot in which particle fracture is absent.

Table 3. Shear strength (τ) of heavily compacted samples of rockfill at low normal stress (σ'), where $\tau = A(\sigma')^b$ and τ and σ' are in kN/m^2

Rock type	A	b
A. Sandstone	6·8	0·67
B1. Slate	5·3	0·75
B2. Slate	3·0	0·77
C. Basalt	4·4	0·81

The plot of $(\sigma_1'/\sigma_3')_f$ against σ_3' to logarithmic scales in Fig. 6 gives a clear indication of the influence of confining pressure on the shear strength of the rockfill. However, the relationship between $(\sigma_1'/\sigma_3')_f$ and σ_3' is not directly useful when selecting shear strength parameters for use in limit equilibrium stability analyses. For this purpose it is simplest to examine the Mohr failure envelopes. The envelopes for the four rockfills are plotted in Fig. 8 for low and medium stresses, up to a normal effective stress of 400 kN/m^2. All four envelopes show marked curvature and there are wide differences between the shear strength of the four materials. Over a limited range of normal stress (40 kN/m^2 $< \sigma' <$ 400 kN/m^2) the shear strength of the rockfills can be described by an equation of the form

$$\tau = A(\sigma')^b \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (2)$$

where A and b are constants for the rockfill.

Table 3 lists the values of A and b for the four rockfills (τ and σ' are in kN/m^2 and b is typically about 0·75). De Mello (1977) has derived values of A and b from Marsal's data. He has shown that values of b (which is independent of the units of τ and σ') range from 0·81 to 0·88. The larger values can be attributed partly to the much higher stress range (σ' up to 5000 kN/m^2) considered by de Mello.

STABILITY OF STRUCTURES BUILT OF COMPACTED ROCKFILL

Rockfill dams, sometimes over 30 m in height, were built in California in the nineteenth century using rockfill dumped loosely in position and with an impervious face next to the water. The slopes of these embankments were often steeper than 1 vertical in 1 horizontal; the construction of slopes steeper than the angle of repose (about 1 in 1·3 usually) was achieved by hand-placing stones to form a rubble-retaining wall (Galloway, 1939). In recent years the adoption of different construction techniques utilizing modern heavy earth-moving plant has led to rockfill being placed in comparatively thin layers and heavily compacted with vibrating rollers. This method results in a very dense, well-graded rockfill with strength and deformation properties greatly superior to those of the loose dumped, uniform sized rockfills previously used. However, as de Mello (1977) has pointed out, modern rockfill embankments are usually built with much flatter slopes than the earlier structures.

Large rockfill embankments are most commonly built to act as dams and in this situation, because of their size and importance as water-retaining structures, their stability and deformation behaviour is usually carefully analysed at the design stage. Where a clay core is incorporated in the embankment to form the watertight element, flatter slopes may be required to maintain stability. For large rockfill embankments (over 100 m high) typical values of ϕ' used in design are 37° to 45° (Lauffer and Schober, 1964; Robeson and Crisp, 1966; Marsal and Ramirez, 1967; Pinkerton and Paton, 1968). These values will probably produce a fairly

realistic factor of safety, although the use of these angles of shearing resistance for small rockfill embankments will be very conservative for some rockfills. Furthermore, the use of a constant angle of shearing resistance in the stability analysis, ignoring the curvature of the failure envelope, will not give the correct position of the critical failure surface.

Many small rockfill embankments are built, particularly in highway engineering, using standardized side slopes and without carrying out any detailed stability analysis (Symons, 1970; Arrowsmith, 1971). Although the British Standard Code of Practice 2003 suggests that side slopes corresponding to the angle of repose of the rockfill in a loose state (typically 1 in 1·3) may be used, in UK highway engineering practice side slopes of 1 vertical in 2 horizontal are generally the steepest adopted for ease of maintenance (Joyce, 1972). When restrictions on available space make such side slopes impractical, retaining walls or reinforced earth construction are often used to reduce the width required for the base of the embankment. For many rockfills the use of realistic shear strength parameters would show that small embankments could be built with very steep side slopes and have an adequate factor of safety against stability failure without any need for reinforcement or for a retaining wall. Special precautions would, however, be needed to contain and protect the surface of side slopes steeper than the angle of repose of loose rockfill. This could be achieved by a layer of fitted or interlocking stones or blocks, the use of gabions, etc.

As it is usually only the stability of large rockfill structures that is analysed, it is understandable that emphasis has been mostly on the testing of rockfill at high confining pressures. However, it has already been pointed out that in a rockfill embankment 50 m high, the normal stress on the critical failure surface is unlikely to exceed 400 kN/m^2. As a consequence, in all but the highest rockfill embankments (the highest rockfill embankment in the UK currently is the 90 m high Llyn Brianne Dam), all the rockfill on a potential failure surface will have a normal stress within the range where the principal stress ratio at failure is strongly influenced by the magnitude of the minor principal stress and the Mohr envelope shows marked curvature. Clearly a realistic stability analysis takes some account of this. For a homogeneous rockfill embankment, the use of shear strength parameters $c' = 0$ and $\phi' =$ a constant implies that surface stability is critical. However, the relationship $\tau = A(\sigma')^b$, where b is about 0·75, indicates that surface stability will not be critical (Charles, 1979).

When a standard method of stability analysis is used, the failure envelope must be defined by values of the parameters c' and ϕ'. The best approximation to the actual curved envelope will be obtained by a straight line such that $c' \neq 0$. An assessment must initially be made of the range of normal stress on the critical failure surface and then values of c' and ϕ' selected to give the best approximation to the actual failure envelope over this stress range. Obviously a more precise analysis could be carried out using shear strength parameters that more accurately describe the curved envelope $\tau = A(\sigma')^b$. Maksimovic (1979) has developed a type of limit equilibrium stability analysis for use with non-linear failure envelopes.

The significance of the curved failure envelope on the depth of the critical failure surface can be assessed by examining translational slides in an infinite slope. For such a slope at an angle to the horizontal of θ, built of rockfill with a bulk density γ and a shear strength $\tau = A(\sigma')^b$, it can be shown that the factor of safety

$$F = \frac{A}{\gamma^{(1-b)}} \frac{\cos^{(2b-1)}\theta}{\sin\theta} \frac{1}{z^{(1-b)}} \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (3)$$

where z is the depth of the slide surface (assuming no pore water pressures). Figure 9 shows a plot of F versus z for a typical good rockfill with $A = 5{\cdot}3$, $b = 0{\cdot}75$, $\gamma = 22$ kN/m^3, and a slope of 1 in 1 ($\theta = 45°$). This approach is valid only for a slope of finite height if z is small compared

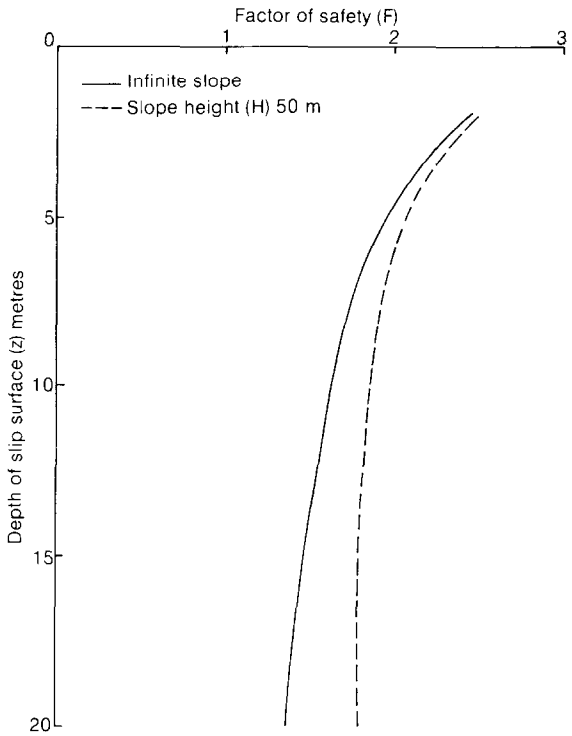


Fig. 9. Stability of 1 in 1 rockfill slope against translational slide ($\tau = 5{\cdot}3\ (\sigma')^{0{\cdot}75}$, where τ and σ' are in kN/m^2; $\gamma_{bulk} = 22$ kN/m^3, no pore water pressures)

to the height of the slope (H). A simplified analysis illustrating the effect of the finite nature of the slope for $H = 50$ m is shown in Fig. 9. The minimum factor of safety (1·8) occurs for a slip surface at a depth of about 17 m. De Mello (1977) concluded that the curvature of the failure envelope did not lead to a significant reduction of F with depth, but the strength data that he used in his analyses were not derived from tests at low stresses and consequently did not result in failure envelopes with such marked curvature.

COMMENTS

The major influence of confining pressure on the principal stress ratio at failure for four rockfills (derived from parent rock materials of very different strengths) has been demonstrated for confining pressures corresponding to the range of stresses likely to be encountered in most rockfill structures. These results have been obtained from drained triaxial compression tests on heavily compacted 38 mm down samples of the rockfills at their natural moisture contents. Consideration needs to be given to the applicability of shear strength parameters derived in this way to actual field situations.

The assumption of drained behaviour in the field should be valid. Rockfill has a high permeability and the field rockfill, having a larger particle size, will usually be much more permeable than the rockfill tested in the laboratory. Penman and Charles (1976) have suggested that a fill described as a rockfill should have a permeability greater than 1×10^{-5} m/s. Heavy compaction on the rockfill at its natural moisture content in the laboratory achieves a reasonable approximation to field density. The use of a triaxial test is largely governed by simplicity and

practical convenience in the laboratory, but the stress conditions in the field situation will often be much closer to plane strain. Tests on rockfill carried out under plane strain conditions (Marsal, 1973) have shown a pattern of behaviour similar to that observed in triaxial tests. However, when the results of tests at the same value of σ_3' are compared, it is found that $(\sigma_1'/\sigma_3')_f$ is significantly greater in the plane strain test. It would seem therefore that the use of shear strength parameters derived from triaxial tests to analyse the stability of rockfill structures is a safe (i.e. conservative) procedure.

Consideration should also be given to the grading of the rockfill that is to be tested in the laboratory. The approach of Lowe (1964) involves the use of a laboratory grading which plots as a line parallel to the field grading on a semi-logarithmic particle size distribution graph. (For a sandstone rockfill Charles, 1973, has shown the geometric similarity of the particles over wide range of particle sizes.) However, the breakdown of particles during compaction makes such an approach difficult to achieve in practice.

An alternative procedure used by Zeller and Wullimann (1957) involved the removal of over-size particles and testing the remaining material. This would seem to have much to commend it as a simple and practical approach. There is no evidence that minor differences in grading have a significant effect on the measured shear strength.

Caution is needed when failure envelopes such as those derived from the triaxial tests described are used for stability analyses. Triaxial compression tests on rockfill have not been carried out at very low confining pressures and so the exact shape of the failure envelope at these low stresses is conjectural. Furthermore, a small positive pore water pressure would drastically reduce the strength of the rockfill at low stress. (Conversely, of course, a suction would greatly increase the strength of the rockfill.) Clearly, rockfill structures should be designed so that positive pore water pressures do not occur. The relationship between $(\sigma_1'/\sigma_3')_f$ and σ_3' is largely a function of dilatancy which is itself strongly dependent on particle shape, strength and degree of compaction. The latter means that adequate compaction in the field must be ensured to obtain the shear strengths measured in the laboratory (Penman and Charles, 1976). A reasonable approach to stability analysis would be to use shear strength parameters representing as closely as possible the failure envelope derived from drained triaxial compression tests, together with a fairly large factor of safety (in the range 1·5 to 2) to allow for some of the uncertainties. The marked drop in shear stress after failure at low confining pressures also makes a conservative factor of safety desirable. Such an approach would be greatly superior to analysis based on an apparently low factor of safety, but with an extra unspecified factor applied to the failure envelope with $c' = 0$ and a constant value for ϕ'.

CONCLUSIONS

The results of drained triaxial compression tests on heavily compacted well-graded samples of a wide range of rockfills all show failure envelopes with pronounced curvature at low stresses, i.e. the value of $(\sigma_1'/\sigma_3')_f$ is strongly dependent on the value of σ_3'.

The maximum principal stress ratio $(\sigma_1'/\sigma_3')_f$ is related to the rate of dilation at failure $(-d\varepsilon_v/d\varepsilon_1)_f$ which is itself a function of particle shape, strength and relative density. A value of $(-d\varepsilon_v/d\varepsilon_1)_f$ greater than 2 has been measured in a test on heavily compacted basalt rockfill at a low confining pressure.

As most stability problems are principally concerned with the strength of rockfill at low normal stress, it is highly desirable to measure the shear strength of rockfill at such low stress levels and to use shear strength parameters that closely approximate to the measured curved failure envelope. This can be described by $\tau = A(\sigma')^b$, where b is likely to be about 0·75 for many heavily compacted rockfills in the low stress range.

The curvature of the failure envelope means that surface stability is not critical even in a homogeneous rockfill embankment. Furthermore, the factor of safety against stability failure will be dependent on the height of the rockfill slope.

Limit equilibrium stability analyses suggest that a 50 m high 1 in 1 slope built with good quality rockfill is likely to have an adequate factor of safety (typically about 1·8). For smaller slope heights steeper slopes could be built and still retain an adequate factor of safety. Where slopes are built at angles steeper than the angle of repose of the loose rockfill, then special measures would be needed to contain and protect the surface of the slope.

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